EE 2700
Digital Circuits

Lab 9 – Garage Door Opener

Objective: In this lab, the student will gain experience by designing a state machine with asynchronous inputs using VHDL and implementing it on an FPGA.

Preparation:

Using the state diagram on the next page, design a controller for a garage door opener in VHDL that has two outputs for the garage door motor, one to drive the door up and the other to drive the door down (both asserted high). There are two stop switches, one that activates when the door is fully open (top) and one that activates when the door is fully closed (bottom). These inputs are asserted (active) low, as is the button input that turns the motor on and off. If the door is closed, the button should make the door go up. If the door is fully open, the button should make the door go down. Otherwise, the door should go in the direction opposite its last movement. Finally, there is a sensor input (also asserted low) that indicates an obstruction. If this signal is asserted while the garage door is going down, the door must immediately (i.e. on the next clock) reverse. Additionally, if the sensor input is asserted when the button is pressed and the motor is stopped, the garage door must be driven upward (unless it is already at the top stop). The clock input is 60Hz. Reset is asserted low.

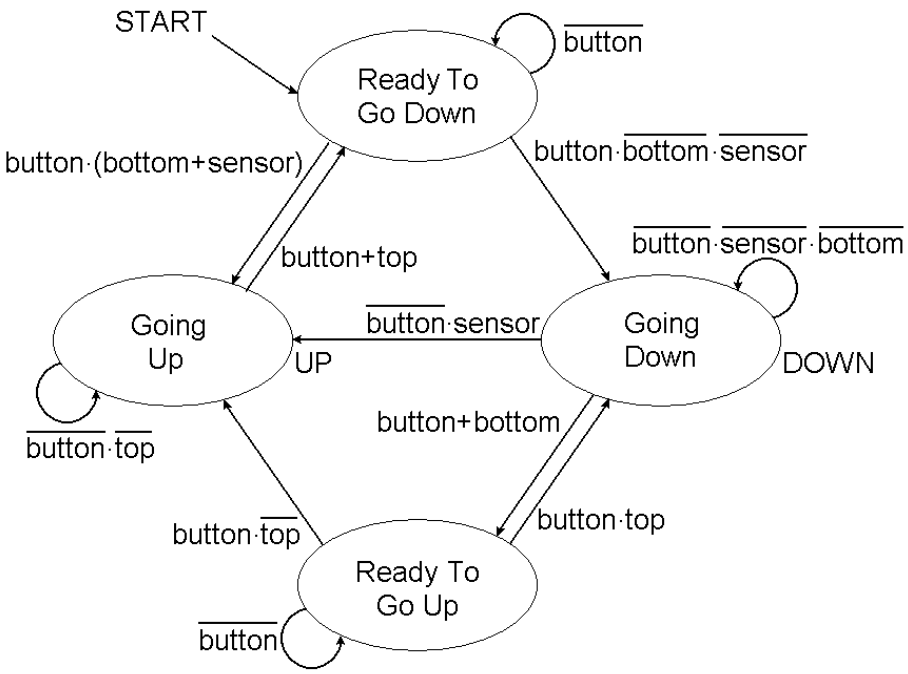
All inputs (except clock and reset) must be synchronized, but the 60Hz clock will make de-bouncing unnecessary. Also, it will be to your advantage to precondition the button signal with a “single pulse” circuit so that it is asserted for one clock period each time the button is pressed.

Design the controller in a single VHDL module to implement the state transition diagram on the next page. Use the entity declaration below:

```
entity garage is
   Port ( RST_L : in  STD_LOGIC;
          TOP_L : in  STD_LOGIC;
          BOTTOM_L : in  STD_LOGIC;
          SENSOR_L : in  STD_LOGIC;
          BUTTON_L : in  STD_LOGIC;
          UP : out  STD_LOGIC;
          DOWN : out  STD_LOGIC;
          CLK : in  STD_LOGIC);
end garage;
```

Write a test bench (store as a simulation source, not a design source) that checks each possible state transition. Simulate your module using Vivado. Save or print the simulation output.

Synthesize and implement your design to verify there are no errors.



Download the files “BasysWrapper.vhd” and “BasysWrapper.xdc” from CANVAS and add them to your project (to Design Sources and Constraints respectively). These files allow you to use Basys switches and LEDs as inputs and outputs to your garage module (in conjunction with the inputs and outputs of the garage door mock-up you will be using in your final test). Study the constraints file to learn which Basys switches are attached to which inputs and record the mapping in your lab book. Repeat the procedure for the Basys LEDs and the connections to actual hardware on PMOD connector JB. Making sure the BasysWrapper is your top level design, synthesize, implement and create a bit file for your garage door opener.

Print your VHDL module (garage.vhd), test bench and simulation output, then affix them to your lab book.

Parts: Basys 3 FPGA Board and Garage Mock-up (Supplied in the lab.)

Procedure: Connect your computer to the Basys 3 board using the microUSB connector. Turn on the Basys 3 board and download your file using the Hardware Manager in Vivado. Exercise the switches and verify the LEDs light as expected (particularly the UP and DOWN LEDs). Once that is working, bring the garage door mock-up to your station and while unplugged, carefully connect it to PMOD connector JB. Plug in the mock-up and verify it functions as required. Demonstrate your working garage door opener to the lab instructor.

Extra Credit (1 point):

Once you have demonstrated your working garage door, write a new constraints file that removes the extra signals used by BasysWrapper.vhd (but retains those used by garage.vhd). Make garage.vhd your top module and rebuild your bit file (using your new constraints file). Download this bit file to your Basys board, connect the garage door mock-up and demonstrate that the garage door still works (but without the Basys LEDs or switches).

Signoff: A lab score can only be given if the circuit is functional.

Rubric (10 points total)

- Lab book contains a title and description and is clearly legible. (1 point)
- Lab book contains a legend of switch, LED and PMOD signals (1 point)
- Lab book contains the VHDL module (garage.vhd) (1 point)
- Lab book contains the test bench (1 point)
- Lab book contains simulation results (1 point)
- Lab book has no obliterations. (1 point)
- Lab book contains a signed, dated summary (1 point)
- Each used page has a page number and is initialed* and dated* (1 point)
- The circuit is functional before the end of the lab period. (2 points)

Note: If the circuit is working at the end of the lab period but the lab book is not yet complete, the lab can be signed off as “working”, and no late penalty will be assessed if it is graded on or before the last day of class.

* It is not necessary to initial and date a page that contains a signature and date unless the dates are different.